Vyoman Jain

Incoming MTD (AI) student at Singapore University of Technology and Design
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WORK EXPERIENCE

Apple | SDE Intern **Jan 2026 – Jun 2026**
Worked on AI/ML systems to automate the end-to-end triage of software issues across Apple technologies including root-cause analysis, deduplication and fix suggestions, as a part of the Wireless Technology and Ecosystems team

CognitiveLab | AI Research Intern **Nov 2024 – Dec 2025**
- Built Nayana, a unified multimodal, multilingual, and multitask AI model family. Developed a custom data generation toolkit for generating synthetic vision datasets in low-resource languages for document understanding tasks
- Produced >2 million pages (2TB+) of data in 22 languages by efficiently scaling the pipeline using Modal, cutting time taken by 90% while also reducing the cost by 50% compared to traditional parallelization with a cluster of VMs
- Built Nayana-OCR, a 500M-parameter fine-tuned vision language model on our synthetic data. It outperformed several contemporary OCR methods, and beat the base model by 50-70% on key metrics across 10 Indic languages
- Developed the M3DR framework to train NetraEmbed and ColNetraEmbed – multilingual visual document retrieval models, which outperform baseline models by 150% on cross-lingual retrieval tasks. Introduced the Nayana-IR benchmark, comprising 23 datasets to evaluate cross-lingual and monolingual visual document retrieval capabilities
- CognitiveLab won the **Llama Impact Grant by Meta** for Nayana – one of the ten recipients globally in 2025

IIIT Bangalore | Research Intern **May 2025 – Jul 2025**
Worked on Language Guided Object Manipulation in 3D Indoor Scenes under Prof. Viswanath G. Finetuned diffusion models for generating high-fidelity synthetic datasets for training Vision-Language-Action and robotics models

IBM Global Remote Mentoring Project | Open-Source Contributor **Jun 2025 – Jul 2025**
Selected by IBM to contribute to zopen community's open-source project on enabling AI inference on z/OS mainframes
- Successfully ported the llama.cpp framework to z/OS, achieving a 92% pass rate on the core test suite and enabling private, on-device AI for enterprise mainframes. Built a PoC to demo practical AI use cases and MCP functionality
- Engineered patches to resolve core compilation and architectural challenges like endianness and threading, integrated acceleration libraries to boost token generation speed and prompt processing by 10%

Sarvam AI | Summer Intern **May 2024 – Aug 2024**
Interned at Sarvam, a sovereign AI pioneer building the GenAI stack for India. Built the core pipeline for SarvamParse – a tool for parsing unstructured data from PDFs into clean, structured data with extremely high accuracy for use in RAG applications.
- Introduced key new features in the LLM-parsing pipeline for robust parsing error handling and boosting parsing accuracy
- Refactored the codebase reducing its size by over 50%, while implementing unit tests, custom logging and optimizing for speed, cost, and efficiency across 90% of all types of documents
- Deployed the project as a scalable Google Cloud service with an API-endpoint and as a Python package

PUBLICATIONS

***HyperCluster**: Decentralized Large Language Model Inference over Peer-to-Peer Wireless Networks* - Poster accepted at ML4Wireless workshop at **AAAI 2026** (non-archival); Full paper accepted at **DAIS 2026**

***ReviewArena**: A Large-Scale Cross-Conference Dataset and Benchmark for LLM Peer Review* - Accepted as **Spotlight** paper at **ICML 2026** AI for Science workshop (non-archival)

***M3DR**: Towards Universal Multilingual Multimodal Document Retrieval* - Accepted at **AAAI 2026** workshop on New Frontiers in Information Retrieval workshop (non-archival)

***Manimator**: Transforming Research Papers into Visual Explanations* - NewInML workshop at **ICML 2025** (poster, non-archival)

***Nayana**: A Foundation for Document-Centric Vision-Language Models via Multi-Task, Multimodal, and Multilingual Data Synthesis* - Poster accepted at **CVPR 2025 Workshop** on Vision Language Models for All (non-archival); Full paper accepted at **ICCV 2025 Workshop** on Computer Vision for Developing Countries

***Nayana OCR**: A Scalable Framework for Document OCR in Low-Resource Languages* - Accepted at **NAACL 2025 Workshop** on Language Models for Underserved Communities

EDUCATION

Master of Science in Technology and Design (Artificial Intelligence), SUTD **Starting Sep 2026**

B. Tech CSE, PES University EC Campus **2022 – 2026**

9.51 CGPA | *Received highest academic scholarship in all semesters (top 5% students), Immense Contribution Award*

GEAR Innovative International School | *Grade XII CBSE – 95.4% | School Vice-Captain* **2022**

Greenwood High International School | *Grade X ICSE – 97.2%* **2020**

SKILLS

- **AI | ML | DL:** PyTorch, sklearn, OpenCV, Transformers, PEFT, llama.cpp, LangChain and LlamaIndex
- **Languages:** Proficient in Python and C; Familiar with Go, C++, JavaScript, R, Java, Shell Scripting and SQL
- **Full Stack Development:** React.js, Next.js, Express.js, Node.js, FastAPI, TailwindCSS, uv, Git
- **Databases:** MongoDB, PostgreSQL, Firebase, MySQL, Supabase, Pinecone, FAISS, Qdrant, LanceDB
- **Cloud:** Docker, Kubernetes, Google Cloud Platform, Microsoft Azure, Amazon Web Services, and Modal for GPUs
- **Big Data** with Hadoop, Kafka, Spark; **Parallel Programming** with OpenMP, CUDA and MPI

PROJECTS

Manimator
A tool to transform research papers and math concepts into stunning visual explanations, powered by AI and the manim engine. Featured in Hugging Face's Spaces of the Week with 5000+ videos generated in a week

Clinical Trial Protocol Document Generator | Alaiy (AI consultancy startup incubated at NSRCEL, IIM-B)
Built a document generation tool for helping medical researchers in easily drafting clinical protocol trial documents for new medicines using agentic RAG for a Singapore-based startup – Neuratrial

Vehicular Troubleshooting Assistance Agent | Bosch University Connect Program
Built a voice-based multilingual Gen AI chatbot using RAG for providing Vehicular Troubleshooting Assistance (VTA) to field technicians, as part of this 6-month project in a collaboration between Centre of Cognitive Computing and Computational Intelligence (C3I) and Bosch. Deployed an agentic version built using LlamaIndex and FastAPI

HyperCluster | Capstone Project
Developed a P2P distributed framework for deploying LLMs on a cluster of edge devices. Implemented a self-organizing P2P network with automatic peer discovery using Iroh. Built a distributed LLM inference layer with model sharding using the HuggingFace transformers library.

EXTRACURRICULARS / ACHIEVEMENTS

GRE Score: 333 / 340 – Valid till 2031

Google Cloud Certified – Associate Cloud Engineer: Certified as Associate Cloud Engineer (ACE) with Google Cloud Skills Boost covering GCP infrastructure skills in Google Compute Engine, Google Kubernetes Engine, Security, Networking, etc.

Subject Matter Expert | PESU I/O: Taught the 'RAGs to Rich AIs with LLM Agents' course to a class of 27 students as part of a university peer-teaching program. Covered a deep dive into Retrieval Augmented Generation (RAG) architecture, advanced RAG techniques, GraphRAG and creating LLM agents with Langchain, LlamaIndex and Crew.ai

Teaching Assistant | Advanced Foundations of Machine Learning: Assisted Dr. Arti Arya in teaching course UE23CS342AA1 to 5th semester AI/ML cohort. Responsible for conducting hands-on sessions and assignments, updating teaching materials and mentoring students for their projects.

Quantum Computing

- Participated in IBM Quantum events to learn and build with Qiskit – IBM Quantum Spring Challenge 2023, IBM Qiskit Global Summer School 2023, IBM Quantum Challenge 2024. Certified as a Qiskit Advocate
- Lead organizer for Qiskit Fall Fest 2024 Extension Event at PES University, selected as one of 100 universities globally to host this IBM Quantum-powered event. Selected again to host an Extension Event for Qiskit Fall Fest 2025, being one of the 150 universities chosen globally from thousands of applicants.

Clubs

- **President, QuantuMania Club PESU ECC**: Led this quantum research community; Conducted workshops, hackathons, lab tours, and guest lectures to foster interest in quantum computing
- **Technical Co-Head, Codechef PESU ECC**: Speaker at workshop to introduce C++ STL to 1st and 2nd year students
- **Tech Domain Member, ACM PESU ECC**: Contributed to 'Hacknight' - a Hacktoberfest event and a workshop on CUDA
- **Quiz Team Member, Quiz Quotient Club PESU ECC**: Organized and set questions for several intra-college quizzes